

# Ion conductive behaviour in a confined nanostructure: NMR observation of self-diffusion in a liquid-crystalline bicontinuous cubic phase†

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**The diffusion of ions in an ionic liquid crystal exhibiting a bicontinuous cubic liquid-crystalline phase has been investigated by NMR spectroscopy in order to examine the behaviour of ions in an ordered nanostructure.**

Nanostructured complex fluids of liquid crystals have attracted a great deal of attention as dynamic functional materials.<sup>1,2</sup> They have been used for the development of functional materials that exhibit low dimensional transport of ions, electrons and molecules.<sup>1a–h,2</sup> Low dimensional ionic conduction has been achieved by using aligned nanostructured ionic liquid crystals,<sup>1b,c,2c</sup> such as smectic and columnar imidazolium salts. However, those nanostructures required macroscopic orientation to function as low dimensional conductors on the macroscale. Recently we found that bicontinuous cubic (Cub<sub>bi</sub>) ionic liquid crystals serve as efficient ion conductors even without having induced macroscopic orientation of the nanostructures.<sup>3</sup> Such phases show great potential for transporting materials.<sup>1h,3</sup> The Cub<sub>bi</sub> liquid crystals have ordered three-dimensional (3D) structures that have been covered in a recent review<sup>4</sup> but, to the best of our knowledge, there has until now been no report on the dynamic properties of thermotropic Cub<sub>bi</sub> phases. Our intention here is to obtain information on ion transport in 3D ordered structures of ionic liquid crystals. A detailed investigation of the molecular self-diffusion in these materials by Pulsed-Field-Gradient Nuclear Magnetic Resonance (PFG-NMR)<sup>5</sup> measurements is a powerful approach to understand the dynamic properties of ions. Herein, we report on the molecular self-diffusion of compound **1** forming a thermotropic Cub<sub>bi</sub> phase (Fig. 1). The advantage of PFG-NMR is its non-invasive nature, molecular selectivity, and the lack of requirement for foreign agents to be added to the sample. This makes PFG-NMR the primary method used for the determination of diffusion

coefficients. Although a number of reports on diffusion coefficients in lyotropic Cub<sub>bi</sub> phases have been presented,<sup>6</sup> no PFG-NMR measurements of functional thermotropic bicontinuous materials have been reported.

Compound **1** has an ammonium salt group and a non-ionic aromatic and aliphatic moiety.<sup>3</sup> Segregation of these incompatible parts in self-assembled states leads to the formation of liquid-crystalline nanostructures.<sup>3</sup> A network of ion channels is formed in the liquid-crystalline phase that promotes a high mobility of the anion, leading to high ion conductivity. At the phase transition from Cub<sub>bi</sub> to isotropic liquid the ionic conductivity decreases by approximately a factor of 4, presumably due to the collapse of the ion channel network. This allows for on/off switching of the ionic conductivity by a temperature induced phase transition.

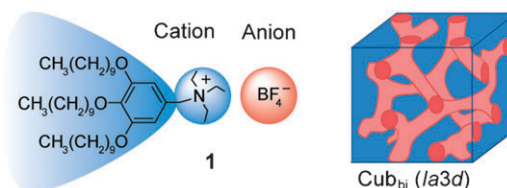
By studying nuclei with different gyromagnetic ratios ( $\gamma$ ), it is straightforward to obtain the distinct self-diffusion coefficients of the different ions in the above mentioned phases. Hence, the self-diffusion coefficients were determined by <sup>1</sup>H- and <sup>19</sup>F-NMR for the cations and anions, respectively. The measurements were performed on a Bruker DMX200 spectrometer equipped with a 10 mm pulsed field gradient probe with 9.6 T/m maximum gradient. The temperature at the sample position was carefully measured with an external thermocouple for both proton and fluorine radiofrequency inserts. A 10 mm NMR tube was filled with 500 mg of the liquid-crystalline material and air bubbles were removed by heating the sample to the isotropic phase, followed by slow cooling to room temperature. For every series of measurements the sample was heated from room temperature with temperature increments of 3–5 K and an equilibration time of 20 min for each temperature step. The stimulated-echo pulse sequence (or similar, see ESI†) was employed and the self-diffusion coefficients were obtained by fitting the echo attenuations to the Stejskal-Tanner equation.<sup>7</sup> The nominal values for all diffusion coefficients were calibrated towards the diffusion of water at 298 K.<sup>8</sup> The self-diffusion data for the

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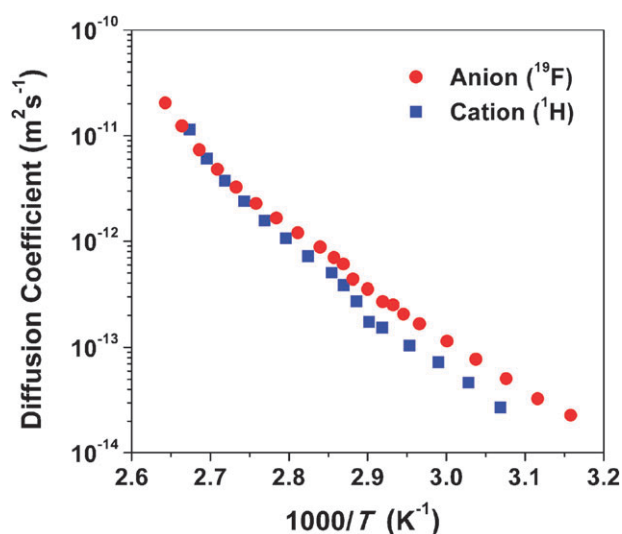
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**Fig. 1** Fan-shaped ammonium tetrafluoroborate salt **1** (left) exhibiting a bicontinuous cubic (Cub<sub>bi</sub>) liquid-crystalline phase (right).



**Fig. 2** Self-diffusion coefficients for anion (red circles) and cation (blue squares).

cation and the anion are presented in Fig. 2. In the  $\text{Cub}_{\text{bi}}$  liquid-crystalline phase the diffusion of the ionic species differ from each other significantly. The anion diffusion is approximately twice as fast as that of the cation throughout the temperature interval investigated. This indicates that the anion can dissociate from the cation and move easily, as would be expected in a network of ion channels. The anion can move from cation to cation in the ordered structure and this explains the previously observed high conductivity in the liquid-crystalline phase.<sup>3</sup> Moreover, we observe that the cationic species have significant diffusion (comparable to that observed in columnar thermotropics<sup>5c</sup>) in the liquid-crystalline phase, even at the lowest temperatures. This indicates that the structure of the ion channel network is of rather dynamic character.

At the phase transition temperature to the isotropic liquid, the diffusion of the anion increases continuously, whereas the diffusion of the cation exhibits a sharp increase to almost the same value as that of the anion. More precisely, the ratio of anion/cation diffusion changes from 2.0 at  $1000/T = 2.9$  to 1.4 at  $1000/T = 2.88$ . Upon further increase of the temperature, the same ratio changes in the same direction but much slower (e.g., becomes 1.3 at  $1000/T = 2.78$ ).

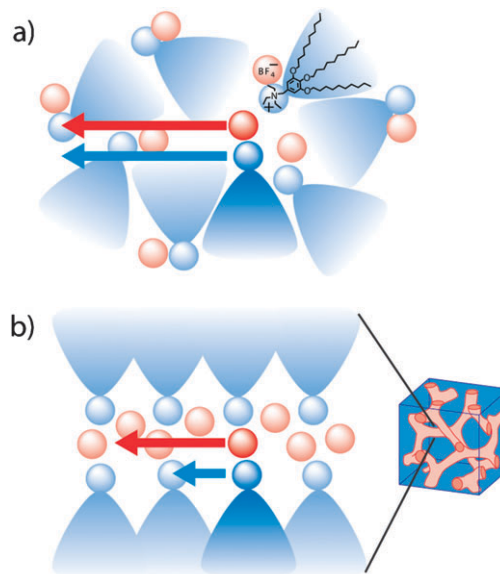
We have further assessed the phase transition temperature from the temperature variation of the longitudinal and transverse relaxation times of the cation signal, commonly known as  $T_1$  and  $T_2$ , (Fig. S1, in ESI†). The relaxation times, sensitive to the local reorientational dynamics, are well documented to be responsive to phase transitions in liquid crystal systems.<sup>9</sup> Although we observe a marked change at around  $1000/T = 2.9$ , the temperature trends in  $T_1$  and  $T_2$  continue to change over a rather broad temperature region up to approximately  $1000/T = 2.8$ . We assume this behaviour is due to the gradual decrease of order in the bicontinuous cubic phase at the temperature range near the isotropization point, as is observed for the decrease of order parameter in a nematic to isotropic transition of a rod-like molecule.<sup>10</sup>

The diffusion coefficients of anion and cation are practically identical in the higher temperature range. This is rather

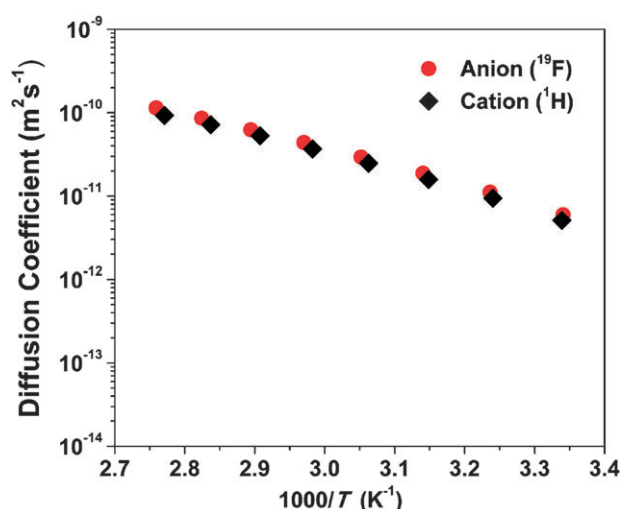
surprising, since the anion, due to its smaller size relative to that of the cation, is expected to show a faster diffusion. We believe that this is the result of formation of ion pairs or ion clusters in the isotropic phase, an idea that has been proposed by several groups before.<sup>11</sup> A proposed molecular motion in isotropic and liquid-crystalline phases is illustrated in Fig. 3. In case of strong ion pairing, the diffusion of the smaller ion becomes controlled by and, in the limiting case, identical to that of the larger ion. Equal diffusion coefficients for the oppositely charged ions should be the case in ionic liquids where the ions are assumed to be always in close proximity to each other. To further support this idea we performed diffusion measurements of a common ionic liquid (Fig. 4) with the same anion (tetrafluoroborate) but a smaller cation (diethyl-methyl-(2-methoxyethyl) ammonium) as compared to compound **1**. The ionic liquid melts at 9 °C and the material lacks any liquid-crystalline phases. The diffusion coefficients of the anion and cation are equal within the error margin of the experiment in the whole temperature range investigated. Moreover, the diffusion coefficients are approximately two orders of magnitude faster compared to those in the isotropic phase of the liquid-crystalline material. This is expected for the cation due to its smaller size compared to the cation of compound **1**. Another contributing factor to the faster diffusion might be less cluster formation in the ionic liquid.

Indeed, the formation of ion pairs in the isotropic phase seems to be the case. Correlation of motion by ion pairing will decrease the concentration of free ions in the material while cluster formation may limit path length for ionic motion. Thereby, these two phenomena can explain the observed decrease of ion conductivity in the isotropic phase.<sup>3</sup>

On the basis of the self-diffusion of the ionic species we have in this work found an explanation for the sudden change in ionic conductivity upon phase transition for an ionic liquid



**Fig. 3** Schematic illustration of the ion self-diffusion in (a) an isotropic phase and (b) a  $\text{Cub}_{\text{bi}}$  liquid-crystalline phase. In (a) the ionic species are observed to form pairs and/or clusters, whereas in (b) a network of ion channels allows for the anions to diffuse separately from the cations.



**Fig. 4** Self-diffusion coefficients for diethyl-methyl-(2-methoxyethyl) ammonium (black diamonds) tetrafluoroborate (red circles).

crystal. The results suggest that dissociating cations and anions in the liquid-crystalline phase form ion pairs and clusters in the isotropic phase, leading to the reduction in the ionic conductivity.

In summary, this communication reports on the use of NMR for the assessment of dynamic function of nanostructured ion-conductive materials. For the first time we have observed the distinct self-diffusion of anions and cations in an ion-conductive material that exhibits a thermotropic bicontinuous cubic liquid-crystalline phase. The knowledge on molecular mobility in nanostructured materials is important for the understanding of their dynamic properties. The results reported here should therefore be valuable for the development of soft materials transporting ions and molecules.

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